

Subconscious Intent Detection v13.0: A 20-Novelty Patent-Grade Real-Time Ocular Micro-Behaviour Classification System Using Stacked Meta-Learning

M. THARUN PRASSANTH K P

School of CSE (SCOPE), VIT Vellore, Tamil Nadu, India

tharun.m2025@vitstudent.ac.in | 25MAI0064 prassanth.kp2025@vitstudent.ac.in | 25MAI0070

Abstract

Subconscious Intent Detection v13.0 is a real-time, webcam-only system that classifies 20 distinct psychological intent states from ocular micro-behaviour signals with zero dependency on proprietary hardware or external libraries. The system is implemented as a single deployable Python file (intent_detection_v13.py) integrating 38 signal channels, 76 derived features, and a six-model stacked generaliser meta-learner ensemble — representing the first application of level-2 meta-learning to webcam-based intent classification in the open literature.

Twenty patent-grade novelties span four domains: (i) **Novel Signal Design** — Inter-Blink Interval Variability (IBIV) capturing blink rhythm irregularity, Peri-Blink EAR Waveform Shape encoding the full EAR trajectory through each blink, Bilateral EAR Phase Coherence measuring left/right eye synchronisation, and Cognitive Momentum Index (CMI) computing the temporal derivative of cognitive load over a 30-frame window; (ii) **Novel Detection Algorithms** — software-only PERCLOS without dedicated hardware, microsleep detection without EEG, and head pose estimation without landmark libraries; (iii) **Novel Architectural Contributions** — Stacked Meta-Learner (N15) with LogisticRegression level-2 learner on out-of-fold predictions, Adaptive Dual-Layer Inference (N5) blending short- and long-window feature vectors, and Online Bayesian Intent Transition Matrix (N18); (iv) **Novel Output Indices** — Eye Strain–Productivity composite index and Personal Baseline Normalisation via 90-frame per-user calibration.

The system features a real-time HUD with 12 live signal bars, cognitive indices, gaze radar, patent signal panel, and Top-3 intent predictions. A five-phase startup sequence ensures personalised operation on any commodity webcam. The stacked meta-learner achieves 100.00% classification accuracy and $F1 = 1.000$ across all 20 intent classes on the held-out synthetic test set derived from 20 published psychophysiological studies.

Index Terms — Subconscious Intent Detection, Eye Aspect Ratio, Blink Analysis, Gaze Entropy, PERCLOS, Stacked Generalisation, Ocular Micro-Behaviour, Cognitive Load, Meta-Learning, IBIV, CMI, Bayesian Transition Matrix, Webcam, Haar Cascade, Random Forest, MLP, Personal Calibration, Patent-Grade Novelty, Real-Time HUD.

I. Introduction

The automatic detection of subconscious psychological states from non-invasive physiological signals has become a central challenge in human-computer interaction, cognitive neuroscience, and safety-critical systems engineering. Ocular micro-behaviour signals — including blink dynamics, gaze distribution, pupil response, and eye aspect ratio — provide a compelling non-invasive pathway accessible through any standard webcam without specialised hardware [1]. Growing availability of commodity imaging devices creates new opportunities to deploy intent detection systems outside laboratory settings, enabling applications from adaptive educational interfaces to cognitive fatigue monitoring in safety-critical environments [12][13].

Despite this opportunity, webcam-based intent inference remains severely limited in scope. Current systems classify at most two intent states — typically drowsy versus alert — and rely on voting ensembles treating all base classifiers as equally informative regardless of the prevailing signal pattern [9][10]. Hardware-dependent approaches using infrared eye-trackers, EEG headsets, or multi-camera rigs have achieved richer intent taxonomies [11], but deployment outside controlled laboratory environments remains impractical for consumer or industrial use.

Four critical limitations define the current state of the art. First, all published systems use soft-vote or hard-vote ensembles assigning uniform weight to each base classifier, ignoring signal-specific reliability [17]. Second, blink rate discards the rhythm variability of inter-blink intervals which Paprocki and Francuz [22] showed to carry independent cognitive information. Third, blink detection is binary — open or closed — losing the full EAR waveform that Abdrabou et al. [7] identified as unexploited. Fourth, cognitive load is estimated as a static instantaneous score with no temporal derivative to identify overload transitions [12].

This paper presents Subconscious Intent Detection v13.0 (SID-v13), a complete real-time system addressing all four limitations through 20 patent-grade novel contributions. SID-v13 classifies 20 psychological intent states in real time from a standard webcam with zero additional hardware. The principal contributions are: (1) a 38-channel signal pipeline with 10 new v13 patent signals; (2) N15 — the first level-2 stacked meta-learner for webcam intent; (3) N11 IBIV — first blink rhythm variability feature; (4) N14 CMI — first cognitive load temporal derivative; (5) N16 PERCLOS without hardware; (6) N17 microsleep without EEG; and (7) complete auto-generating output pipeline producing accuracy JSON, PDF report, and publication-quality graphs on every session exit.

The system requires only standard open-source packages (opencv-python, scikit-learn, pandas, matplotlib, joblib, numpy, scipy, fpdf2) and operates with no cloud connectivity, paid SDKs, or proprietary APIs. It deploys from a single terminal command on any consumer laptop. Model training completes in 3–5 minutes on first run; subsequent runs load persisted models in under 5 seconds.